

Aggregate Functions

1. Count

Code:

```
SELECT COUNT(*) AS total_students FROM students;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| students              |
+-----+
+-----+
| total_students |
+-----+
|          8 |
+-----+
```

2. Sum

Code:

```
SELECT SUM(marks) AS total_marks FROM students;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| students              |
+-----+
+-----+
| total_marks |
+-----+
|          685 |
+-----+
```

3. Avg

Code:

```
SELECT AVG(marks) AS average_marks FROM students;
```

Output:

+-----+	
Tables_in_sandbox_db	
+-----+	
students	
+-----+	
+-----+	
average_marks	
+-----+	
85.6250	
+-----+	

4. Max

Code:

```
SELECT MAX(marks) AS highest_marks FROM students;
```

Output:

+-----+	
Tables_in_sandbox_db	
+-----+	
students	
+-----+	
+-----+	
highest_marks	
+-----+	
95	
+-----+	

5. Min

Code:

```
SELECT MIN(marks) AS lowest_marks FROM students;
```

Output:

Tables_in_sandbox_db	
students	
lowest_marks	
75	

6. Group By

Code:

```
SELECT department, AVG(marks) AS average_marks FROM students  
GROUP BY department;
```

Output:

student_id	name	department	age	marks
1	Arun	CSE	20	85
2	Priya	CSE	21	92
3	Rahul	ECE	20	78
4	Divya	ECE	22	88
5	Karthik	CSE	20	75
6	Meena	IT	21	95
7	Vijay	IT	22	82
8	Anjali	ECE	20	90
department		average_marks		
CSE		84.0000		
ECE		85.3333		
IT		88.5000		

7. Having

Code:

```
SELECT department, AVG(marks) AS average_marks FROM students
GROUP BY department
HAVING AVG(marks) > 85;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| students              |
+-----+

+-----+-----+-----+-----+-----+
| student_id | name   | department | age | marks |
+-----+-----+-----+-----+-----+
| 1 | Arun   | CSE        | 20  | 85    |
| 2 | Priya  | CSE        | 21  | 92    |
| 3 | Rahul  | ECE        | 20  | 78    |
| 4 | Divya  | ECE        | 22  | 88    |
| 5 | Karthik | CSE        | 20  | 75    |
| 6 | Meena  | IT         | 21  | 95    |
| 7 | Vijay  | IT         | 22  | 82    |
| 8 | Anjali | ECE        | 20  | 90    |
+-----+-----+-----+-----+-----+

+-----+-----+
| department | average_marks |
+-----+-----+
| ECE        | 85.3333      |
| IT         | 88.5000      |
+-----+-----+
```

8. Order By

Code:

```
SELECT department, AVG(marks) AS average_marks FROM students  
GROUP BY department  
ORDER BY average_marks DESC;
```

Output:

+-----+				
Tables_in_sandbox_db				
+-----+				
students				
+-----+				
+-----+-----+-----+-----+-----+				
student_id	name	department	age	marks
+-----+-----+-----+-----+-----+				
1	Arun	CSE	20	85
2	Priya	CSE	21	92
3	Rahul	ECE	20	78
4	Divya	ECE	22	88
5	Karthik	CSE	20	75
6	Meena	IT	21	95
7	Vijay	IT	22	82
8	Anjali	ECE	20	90
+-----+-----+-----+-----+-----+				
+-----+-----+				
department	average_marks			
+-----+-----+				
IT	88.5000			
ECE	85.3333			
CSE	84.0000			
+-----+-----+				